

HP-Ganciclovir 500 mg powder for concentrate for solution for infusion

Composition

One vial contains 543 mg ganciclovir sodium, equivalent to 500 mg ganciclovir.

After reconstitution in 10 ml water for injections, 1 ml of the reconstituted solution contains 50 mg of ganciclovir.

Product Description

Powder for concentrate for solution for infusion.

White porous cake or powder.

Reconstituted solutions of ganciclovir should be clear and colorless.

Further diluted solution of ganciclovir should be clear and colorless.

Pharmacodynamics

Mechanism of Action

Ganciclovir is a synthetic analogue of 2'-deoxyguanosine which inhibits replication of herpes viruses in vitro and in vivo. Sensitive human viruses include human cytomegalovirus (HCMV), herpes-simplex virus-1 and -2 (HSV-1 and HSV-2), human herpesvirus -6, -7 and -8 (HHV-6, HHV-7, HHV-8), Epstein-Barr virus (EBV), varicella-zoster virus (VZV) and hepatitis B virus. However, clinical studies have been limited to assessment of efficacy in patients with CMV infection.

In CMV infected cells ganciclovir is initially phosphorylated to ganciclovir monophosphate by the viral protein kinase, UL97. Further phosphorylation occurs by several cellular kinases to produce ganciclovir triphosphate, which is then slowly metabolised intracellularly. This has been shown to occur in HSV- and HCMV infected cells with half-lives of 18 and between 6 and 24 hours respectively after removal of extracellular ganciclovir. As the phosphorylation is largely dependent on the viral kinase, phosphorylation of ganciclovir occurs preferentially in virus-infected cells.

The virustatic activity of ganciclovir is due to the inhibition of viral DNA synthesis by: (1) competitive inhibition of incorporation of deoxyguanosine triphosphate into DNA by DNA polymerase, and (2) incorporation of ganciclovir triphosphate into viral DNA causing termination of, or very limited, viral DNA elongation.

Typical anti-viral IC₅₀ against CMV in vitro is in the range 0.08 µM (0.02 µg/mL) to 14 µM (3.5 µg/mL).

Clinical/Efficacy Studies

No text on clinical studies.

Viral resistance

Viruses resistant to ganciclovir can arise after chronic dosing with ganciclovir or valganciclovir by selection of mutations in either the viral kinase gene (UL97) responsible for ganciclovir monophosphorylation or the viral polymerase gene (UL54). UL97 mutations arise earlier and more frequently than mutations in UL54. Virus containing mutations in the UL97 gene is resistant to ganciclovir alone, with M460V/I, H520Q, C592G, A594V, L595S, C603W being the most frequently reported ganciclovir resistance-associated substitutions. Mutations in the UL54 gene may show cross-resistance to other antivirals targeting the viral polymerase, and vice versa [106, 107, 108]. Amino acid substitutions in UL54 conferring cross-resistance to ganciclovir and cidofovir are generally located within the exonuclease domains and region V,

however amino acid substitutions conferring cross-resistance to foscarnet are diverse, but concentrate at and between regions II (codon 696-742) and III (codon 805-845). The possibility of viral resistance should be considered in patients who repeatedly show poor clinical response or experience continuous viral excretion during treatment.

Pharmacokinetics

The pharmacokinetics of IV ganciclovir is linear over the range of 1.6-5.0 mg/kg.

The systemic exposure (AUC₀₋₂₄) reported following dosing with a single 1 hour IV infusion of 5 mg/kg ganciclovir in HIV+/CMV+ patients or in adult AIDS patients ranged from 18.8 to 26.0 µg.h/mL. In this patient population peak plasma concentration (C_{max}) ranged from 7.59 to 9.03 µg/mL.

Absorption

Not applicable.

Distribution

For IV ganciclovir, the volume of distribution is correlated with body weight with values for the steady state volume of distribution ranging from 0.54 to 0.87 L/kg. Ganciclovir penetrates the cerebrospinal fluid and diffuses across the placenta. Binding to plasma proteins was 1-2% over ganciclovir concentrations of 0.5 and 51 µg/mL.

Metabolism

Ganciclovir is not metabolized to a significant extent.

Elimination

Renal excretion of unchanged drug by glomerular filtration and active tubular secretion is the major route of elimination of ganciclovir. In patients with normal renal function, greater than 90% of IV administered ganciclovir was recovered unmetabolized in the urine within 24 hours. In patients with normal renal function, systemic clearance ranged from 2.64 ± 0.38 mL/min/kg (N=15) to 4.52 ± 2.79 mL/min/kg (N=6) and renal clearance ranged from 2.57 ± 0.69 mL/min/kg (N=15) to 3.48 ± 0.68 mL/min/kg (N=20), corresponding to 90-101% of administered ganciclovir. Half-lives in patients without renal impairment ranged from 2.73 ± 1.29 (N=6) to 3.98 ± 1.78 hours (N=8).

Pharmacokinetics in Special Populations

Pediatric Population

Ganciclovir pharmacokinetics were studied in 27 neonates aged 2 to 49 days at intravenous doses of 4 mg/kg (N=14) and 6 mg/kg (N=13). Mean C_{max} was 5.5 ± 6 µg/mL and 7.0 ± 1.6 µg/mL the lower and higher dose levels respectively. Mean values for V_{ss} (0.7 L/kg) and systemic clearance (3.15 ± 0.47 mL/min/kg at 4 mg/kg and 3.55 ± 0.35 mL/min/kg at 6 mg/kg) were comparable to those observed in adults with normal renal function. Ganciclovir pharmacokinetics were also studied in 10 children with normal renal function, aged 9 months to 12 years. The pharmacokinetic characteristics of ganciclovir were the same after single and multiple (q12h) IV doses (5 mg/kg). Exposure as measured by mean AUC_∞ on days 1 and 14 were 19.4 ± 7.1 and 24.1 ± 14.6 µg.h/mL respectively and the corresponding C_{max} values were 7.59 ± 3.21 (day 1) and 8.31 ± 4.9 µg/mL (day 14). These range of exposures are comparable to those observed in adults. The steady state volume of distribution after a single dose on day 1 and at the end of the repeat dose period (day 14) was 0.68 ± 0.20 L/kg. Systemic clearance for the same study days was 4.66 ± 1.72 (day 1) and 4.86 ± 2.96 mL/min/kg (day 14). The respective

mean values for renal clearance (0 – 12 h) were 3.49 ± 2.40 on day 1 and 3.49 ± 1.19 mL/min/kg on day 14. The corresponding mean values for the half-life were 2.49 ± 0.57 (day 1) and 2.22 ± 0.76 h (day 14). The pharmacokinetics of ganciclovir from this study were consistent with those in neonates and adults.

Geriatric population

No ganciclovir pharmacokinetic studies have been conducted in adults older than 65 years of age. However, because ganciclovir is mainly renally excreted and since renal clearance decreases with age a decrease in ganciclovir total body clearance and prolongation of ganciclovir elimination half-life can be anticipated in the elderly (see section Recommended Dosage, Special dosage instructions, Geriatric patients).

Renal impairment

The total body clearance of ganciclovir is linearly correlated with creatinine clearance. In patients with mild, moderate, and severe renal impairment, mean systemic clearances of 2.1, 1.0 and 0.3 mL/min/kg were observed. Patients with renal impairment show an increased elimination half-life. In patients with severe renal impairment elimination half-life was increased by 10-fold (see section Recommend Dosage, Special dosage instructions, Renal impairment).

Patients undergoing hemodialysis

Plasma concentrations of ganciclovir are reduced by about 50% during a 4-hour hemodialysis session (see section Symptoms and Treatment of Overdose). During intermittent hemodialysis, estimates for the clearance of ganciclovir ranged from 42 to 92 mL/min, resulting in intra-dialytic half-lives of 3.3 to 4.5 hours. Estimates of ganciclovir clearance for continuous dialysis were lower but resulted in greater removal of ganciclovir over a dose interval. For intermittent hemodialysis, the fraction of ganciclovir removed in a single dialysis session varied from 50-63%.

Hepatic impairment

No pharmacokinetic study has been conducted and no population PK data were collected in patients with hepatic impairment undergoing ganciclovir therapy. Hepatic impairment is not anticipated to affect the pharmacokinetics of ganciclovir since ganciclovir is excreted renally (see section Pharmacokinetics, Elimination).

Indication

Cytomegalovirus retinitis in immunocompromised individuals including patients with acquired immunodeficiency syndrome (AIDS). Prevention of CMV disease in transplant patients at risk for CMV disease. Prevention and treatment of life- or sight-threatening cytomegalovirus (CMV) disease in immunocompromised individuals.

Recommended Dosage

General

Caution: do not administer by rapid or bolus IV injection! The toxicity of Ganciclovir may be increased as a result of excessive plasma levels.

Caution: i.m. or s.c. injection may result in severe tissue irritation due to the high pH (~11) of ganciclovir solutions.

The recommended dosage, frequency, or infusion rates should not be exceeded. See Section Instruction for Use, for details of preparation and handling of Ganciclovir solution.

Standard dosage for treatment of CMV retinitis

Induction treatment: 5 mg/kg given as an IV infusion over one hour, every 12 hours for 14-21 days in patients with normal renal function.

Maintenance treatment: 5 mg/kg given as an IV infusion over one hour, once daily on 7 days per week or 6 mg/kg once daily on 5 days per week.

Standard dosage for prevention in transplant recipients

Induction treatment: 5 mg/kg given as an IV infusion over one hour, every 12 hours for 7-14 days in patients with normal renal function.

Maintenance treatment: 5 mg/kg given as an IV infusion over one hour, once daily on 7 days per week or 6 mg/kg once daily on 5 days per week.

Special dosage instructions

Pediatric patients

Safety and efficacy of ganciclovir in pediatrics have not been established, including use for the treatment of congenital or neonatal CMV infections. The use of Ganciclovir in children warrants extreme caution due to the potential for long-term carcinogenicity and reproductive toxicity. The benefits of treatment should outweigh the risks.

Geriatric patients

No studies have been conducted in adults older than 65 years of age. Since renal clearance decreases with age, Ganciclovir should be administered to geriatric patients with special consideration of their renal status.

Patients with renal impairment

For patients with renal impairment, the dose of Ganciclovir should be modified as shown in the table below.

Table 1-Ganciclovir dosing for patients with renal impairment receiving mg/kg dosing:

CrCl	Induction dose	Maintenance dose
≥70 ml/min	5.0 mg/kg q12h	5.0 mg/kg/day
50-69 ml/min	2.5 mg/kg q12h	2.5 mg/kg/day
25-49 ml/min	2.5 mg/kg/day	1.25 mg/kg/day
10-24 ml/min	1.25 mg/kg/day	0.625 mg/kg/day
<10 ml/min	1.25 mg/kg 3x/wk after hemodialysis	0.625 mg/kg 3x/wk after hemodialysis

Estimated creatinine clearance can be related to serum creatinine by the following formulae:

For males:
$$\frac{(140 - \text{age}[\text{years}]) \times (\text{body weight}[\text{kg}])}{(72) \times (0.011 \times \text{serum creatinine}[\mu\text{mol/L}])}$$

For females: 0.85 x male value

As dosage modifications are recommended in patients with renal impairment, serum creatinine or estimated creatinine-clearance levels should be monitored carefully.

Hepatic impairment

The safety and efficacy of Ganciclovir have not been studied in patients with hepatic impairment.

Route of Administration

For Intravenous Infusion

Contraindications

Ganciclovir is contraindicated in patients with hypersensitivity to ganciclovir, valganciclovir or to any of the excipients.

Breastfeeding (see section Pregnancy and Lactation, Lactation).

Warnings and Precautions

General

Cross-hypersensitivity

Due to the similarity of the chemical structure of ganciclovir and that of aciclovir and penciclovir, a cross-hypersensitivity reaction between these drugs is possible. Caution should therefore be used when prescribing Ganciclovir to patients with known hypersensitivity to aciclovir or penciclovir (or to their prodrugs, valaciclovir or famciclovir respectively).

Mutagenicity, teratogenicity, carcinogenicity, fertility, and contraception

Ganciclovir was found to be mutagenic, teratogenic, carcinogenic and to impair fertility. Ganciclovir should therefore be considered a potential teratogen and carcinogen in humans with the potential to cause birth defects and cancers. Prior to initiation of ganciclovir treatment, patients should be advised of the potential risks to the fetus and to use contraceptive measures. Ganciclovir may cause temporary or permanent inhibition of spermatogenesis in males.

Myelosuppression

Ganciclovir should be used with caution in patients with pre-existing hematological cytopenia or a history of drug-related hematological cytopenia and in patients receiving radiotherapy. Severe leukopenia, neutropenia, anemia, thrombocytopenia, pancytopenia, bone marrow failure and aplastic anemia have been observed in patients treated with Ganciclovir. Therapy should not be initiated if the absolute neutrophil count is less than 500 cells/ μ L or the platelet count is less than 25,000 cells/ μ L or the hemoglobin is less than 8 g/dL. It is recommended that complete blood counts and platelet counts be monitored in all patients during therapy, particularly in patients with renal impairment.

In patients with severe leukopenia, neutropenia, anemia and/or thrombocytopenia, that treatment with hematopoietic growth factors and/or the interruption of therapy is recommended.

Use with other medicines

Seizures have been reported in patients taking imipenem-cilastatin and ganciclovir. Ganciclovir should not be used concomitantly with imipenem-cilastatin unless the potential benefits outweigh the potential risks.

Zidovudine and Ganciclovir each have the potential to cause neutropenia and anemia. Some patients may not tolerate concomitant therapy at full dosage.

Didanosine plasma concentrations may increase during concomitant use with Ganciclovir; therefore, patients should be closely monitored for didanosine toxicity.

Concomitant use of other drugs that are known to be myelosuppressive or associated with renal impairment with Ganciclovir may result in added toxicity.

Drug Abuse and Dependence

No information is available for drug abuse and dependence with ganciclovir.

Interactions with Other Medication

Imipenem-cilastatin

Convulsions have been reported in patients who received ganciclovir and imipenem-cilastatin and a pharmacodynamic interaction between these two drugs cannot be discounted. These drugs should not be used concomitantly unless the potential benefits outweigh the risks.

Potential drug interactions

Toxicity may be enhanced when ganciclovir is co-administered with other drugs known to be myelosuppressive or associated with renal impairment. This includes nucleoside analogues (e.g. zidovudine, didanosine, stavudine), immunosuppressants (e.g. ciclosporin, tacrolimus, mycophenolate mofetil), antineoplastic agents (e.g. doxorubicin, vincristine, vinblastine, hydroxyurea), and anti-infectives (e.g. trimethoprim/sulphonamides, dapsone, amphotericin B, flucytosine, pentamidine). Therefore, these drugs should only be considered for concomitant use with ganciclovir if the potential benefits outweigh the potential risks.

Zidovudine

Both zidovudine and Ganciclovir have the potential to cause neutropenia and anemia, a pharmacodynamic interaction may occur during concomitant administration of these drugs some patients may not tolerate concomitant therapy at full dosage.

Didanosine

Didanosine plasma concentrations were found to be consistently raised when given with IV ganciclovir. At intravenous doses of 5 and 10 mg/kg/day, an increase in the AUC of didanosine ranging from 38 to 67% has been observed confirming a pharmacokinetic interaction during the concomitant administration of these drugs. There was no significant effect on ganciclovir concentrations. Patients should be closely monitored for didanosine toxicity (e.g. pancreatitis)

Probenecid

Probenecid given with oral ganciclovir resulted in statistically decreased renal clearance of ganciclovir leading to statistically significant increased exposure. These changes were consistent with a mechanism of interaction involving competition for renal tubular excretion. Therefore, patients taking probenecid and Ganciclovir should be closely monitored for ganciclovir toxicity.

Pregnancy and Lactation

Fertility

Ganciclovir was found to impair fertility.

Contraception

Women of reproductive potential should be advised to use effective contraception during and for at least 30 days after treatment. Sexually active men are recommended to use condoms during and for at least 90 days after cessation of treatment with Ganciclovir, unless it is certain that the female partner is not at risk of becoming pregnant.

Pregnancy

Ganciclovir was associated with reproductive toxicity and teratogenicity.

The safety of Ganciclovir in pregnant women has not been established. However, ganciclovir readily diffuses across the human placenta. The use of Ganciclovir should be avoided in pregnant women unless the benefit to the mother outweighs the potential risk to the fetus. The safe use of Ganciclovir during labor and delivery has not been established.

Lactation

Peri- and postnatal development has not been studied with ganciclovir but the possibility of ganciclovir being excreted in breast milk and causing serious adverse reactions in the nursing infant cannot be discounted. Human data are not available, but animal data indicates that ganciclovir is excreted in the milk of lactating rats. Therefore, a decision should be made to discontinue the drug or discontinue nursing taking into consideration the potential benefit of Ganciclovir to the nursing mother.

Side Effects

Valganciclovir is a pro-drug of ganciclovir, and adverse reactions associated with valganciclovir can be expected to occur with ganciclovir. Therefore, adverse drug reactions reported with IV or oral ganciclovir (no longer available) or with valganciclovir are included in the table of adverse reactions.

In patients treated with ganciclovir/valganciclovir the most serious and frequent adverse drug reactions are hematological reactions and include neutropenia, anemia and thrombocytopenia.

The frequencies presented in the table of adverse reactions are derived from a pooled population of HIV-infected patients receiving maintenance therapy with ganciclovir or valganciclovir. Exception is made for agranulocytosis, granulocytopenia and anaphylactic reaction; the frequencies of which are derived from post-marketing experience.

Frequencies are presented as CIOMS frequency categories defined as very common, common, uncommon, rare and very rare.

The overall safety profile of ganciclovir/valganciclovir is consistent in HIV and transplant populations except that retinal detachment has only been reported in HIV patients with CMV retinitis. However, there are some differences in the frequency of certain reactions.

Intravenous ganciclovir is associated with a lower risk of diarrhea compared to oral valganciclovir. Pyrexia, candida infections, depression, severe neutropenia (ANC <500 μ L) and skin reactions are reported more frequently in patients with HIV. Renal and hepatic dysfunction is reported more frequently in organ transplant recipients.

Table-2 Frequency of Ganciclovir/Valganciclovir ADRs Reported in HIV Patients Receiving Maintenance Therapy

ADR (MedDRA) System Organ Class	Frequency Category
<i>Infections and infestations:</i>	
Candida infections including oral candidiasis	Very common
Upper respiratory tract infection	
Sepsis	Common
Influenza	
Urinary tract infection	
Cellulitis	
<i>Blood and lymphatic system disorders:</i>	
Neutropenia	Very common
Anemia	
Thrombocytopenia	Common
Leukopenia	
Pancytopenia	
Bone marrow failure	Uncommon
Aplastic anemia	Rare
Agranulocytosis*	
Granulocytopenia*	
<i>Immune system disorders:</i>	
Hypersensitivity	Common
Anaphylactic reaction*	Rare
<i>Metabolism and nutrition disorders:</i>	
Decreased appetite	Very common
Weight decreased	Common
<i>Psychiatric disorders:</i>	
Depression	Common
Confusional state	
Anxiety	
Agitation	Uncommon
Psychotic disorder	
Thinking abnormal	
Hallucinations	
<i>Nervous system disorders:</i>	
Headache	Very common
Insomnia	Common
Neuropathy peripheral	
Dizziness	
Paraesthesia	
Hypoaesthesia	

Seizures	Uncommon
Dysgeusia (taste disturbance)	
Tremor	
Eye disorders:	
Visual impairment	Common
Retinal detachment**	
Vitreous floaters	
Eye pain	
Conjunctivitis	
Macular edema	
Ear and labyrinth disorders:	
Ear pain	Common
Deafness	Uncommon
Cardiac disorders:	
Arrhythmia	Uncommon
Vascular disorders:	
Hypotension	Common
Respiratory, thoracic and mediastinal disorders:	
Cough	Very common
Dyspnoea	
Gastrointestinal disorders:	
Diarrhea	Very common
Nausea	
Vomiting	
Abdominal pain	
Dyspepsia	Common
Flatulence	
Abdominal pain upper	
Constipation	
Mouth ulceration	
Dysphagia	
Abdominal distention	
Pancreatitis	
Hepatobiliary disorders:	
Blood alkaline phosphatase increased	Common
Hepatic function abnormal	
Aspartate aminotransferase increased	
Alanine aminotransferase increased	
Skin and subcutaneous tissue disorders:	
Dermatitis	Very common
Night sweats	Common

Pruritus	Uncommon
Rash	
Alopecia	
Dry skin	
Urticaria	
<i>Musculoskeletal and connective tissue disorders:</i>	
Back pain	Common
Myalgia	
Arthralgia	
Muscle spasms	
<i>Renal and urinary disorders:</i>	
Renal impairment	Common
Creatinine clearance renal decreased	
Blood creatinine increased	
Renal failure	Uncommon
Hematuria	
<i>Reproductive system and breast disorders:</i>	
Infertility male	Uncommon
<i>General disorders and administration site conditions:</i>	
Pyrexia	Very common
Fatigue	
Injection site reaction	Common
Pain	
Chills	
Malaise	
Asthenia	
Chest pain	
	Uncommon

* The frequencies of these adverse reactions are derived from post-marketing experience.

** Retinal detachment has only been reported in studies in HIV patients treated with Ganciclovir for CMV retinitis.

Description of selected adverse reactions

Neutropenia

The risk of neutropenia is not predictable on the basis of the number of neutrophils before treatment. Neutropenia usually occurs during the first or second week of induction therapy. The cell count usually normalizes within 2 to 5 days after discontinuation of the drug or dose reduction.

Thrombocytopenia

Patients with low baseline platelet counts ($< 100,000 \mu/L$) have an increased risk of developing thrombocytopenia. Patients with iatrogenic immunosuppression due to treatment with immunosuppressive drugs are at greater risk of thrombocytopenia than patients with HIV. Severe thrombocytopenia may be associated with potentially life-threatening bleeding.

Laboratory Abnormalities

Laboratory abnormalities in HIV infected patients

Laboratory abnormalities reported from three clinical trials in HIV infected patients receiving intravenous ganciclovir as maintenance treatment for CMV retinitis are listed below in Table 3. One hundred seventy-nine patients were eligible for the laboratory abnormality analysis.

Table 3 Laboratory abnormalities:

Laboratory abnormalities	N=179
Neutropenia (ANC /mm ³) <500 500–<750 750–<1000	25.1% 14.3% 26.3%
Anemia (hemoglobin g/dl) <6.5 6.5–<8.0 8.0–<9.5	4.6% 16.0% 25.7%
Thrombocytopenia (platelets/mm ³) <25,000 25,000–<50,000 50,000–<100,000	2.9% 5.1% 22.9%
Serum creatinine (mg/dl) >2.5 >1.5–2.5	1.7% 13.9%

Post-marketing Experience

Safety reports from the post-marketing setting are consistent with safety data from clinical trials with ganciclovir and valganciclovir.

Symptoms and Treatment of Overdose

Overdose experience with intravenous ganciclovir

Reports of overdoses with intravenous ganciclovir, some with fatal outcomes, have been received from clinical trials and during post-marketing experience. In some of these cases no adverse events were reported. The majority of patients experienced one or more of the following adverse events:

- *Hematological toxicity*: myelosuppression including pancytopenia, bone marrow failure, leukopenia, neutropenia, granulocytopenia
- *Hepatotoxicity*: hepatitis, liver function disorder
- *Renal toxicity*: worsening of hematuria in a patient with pre-existing renal impairment, acute kidney injury, elevated creatinine
- *Gastrointestinal toxicity*: abdominal pain, diarrhea, vomiting
- *Neurotoxicity*: generalised tremor, seizure

Hemodialysis and hydration may be of benefit in reducing blood plasma levels in patients who receive an overdose of ganciclovir.

Effects on Ability to Drive and Use Machines

No studies on the effect on the ability to drive and use machines have been performed. Based on the adverse reaction profile, ganciclovir may have a minor influence on the ability to drive and use machines. Adverse reactions, for example seizures, dizziness and confusion may occur in patients receiving Ganciclovir. If they occur, such effects may affect tasks requiring alertness including the patient's ability to drive and operate machinery.

Instruction for Use

Caution should be exercised in the handling of Ganciclovir. Since Ganciclovir is considered a potential teratogen and carcinogen in humans, caution should be observed in its handling.

Avoid inhalation or direct contact of the powder contained in the vials or direct contact of the reconstituted solution with the skin or mucous membranes. Ganciclovir solutions are alkaline (pH ~11). If such contact occurs, wash thoroughly with soap and water, rinse eyes thoroughly with plain water.

Wearing disposable gloves is recommended during reconstitution and when wiping the outer surface of the vials and the table after reconstitution.

Incompatibilities

Ganciclovir should not be mixed with other IV products.

Preparation of Ganciclovir reconstituted solution

1. Lyophilized Ganciclovir should be reconstituted by injecting 10 mL of sterile water for injection into the vial. Do not use bacteriostatic water for injection containing parabens (parahydroxybenzoates), since these are incompatible with Ganciclovir sterile powder and may cause precipitation.

2. The vial should be gently swirled in order to ensure complete wetting of the product. Continue swirling until a clear reconstituted solution is obtained. Reconstituted solution should be inspected for particulate matter prior to proceeding with admixture preparation.

3. From a microbiological point of view, the reconstituted solution should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Preparation of Ganciclovir infusion solution

Based on patient weight the appropriate calculated dose volume should be removed from the Ganciclovir vial (concentration 50 mg/mL) and added to an acceptable infusion fluid (typically 100ml). Normal saline, dextrose 5% in water, Ringer's or lactated Ringer's solution are determined chemically or physically compatible with Ganciclovir. Infusion concentrations greater than 10 mg/mL are not recommended.

From a microbiological perspective, because Ganciclovir is reconstituted with non-bacteriostatic sterile water, the infusion solution should be used as soon as possible. If not used immediately in-use storage times and conditions prior to use are the responsibility of the user and should not be longer than 24 hours at 2°C to 8°C.

Disposal of unused/expired medicines

The release of pharmaceuticals in the environment should be minimized.

Medicines should not be disposed of via wastewater and disposal through household waste should be avoided.

The following points should be strictly adhered to regarding the use and disposal of syringes and other medicinal sharps:

- Needles and syringes should never be reused.
- Place all used needles and syringes into a sharp's container (puncture-proof disposable container).
- Dispose of the full container and of the administration system according to local requirements.

Storage condition

Storage

Do not store above 30°C.

Storage of the reconstituted solution

For immediate use after reconstitution.

Storage of the infusion solution

Chemical and physical in-use stability of the infusion solution has been demonstrated for 24 hours at 2–8°C. Do not freeze.

Dosage Form and Packaging Available

Carton with 1,5 or 10 colorless and Transparent Type I glass vials of 10ml with a rubber stopper and aluminum-plastic overseal (aluminium crimp with a plastic flip-off top), containing 500mg ganciclovir in form of sterile lyophilized ganciclovir sodium.

Manufacturer

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